

VPL: A Novel Cost Model for Earth and Mars-Orbit Navigation and Communication Constellations

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The problem

Limited public unit-cost data for navigation and communication constellations. Legacy models (SMAD USCM8, QuickCost) rest on data up to the early 2000s. We build a cost model on **27 systems** (1997-2025), that also cover the New-Space era.

The model

$$\text{TOTAL}(M, N) = F g \left[\underbrace{3.29 M^{0.682}}_{\text{DEV}(M)} + \underbrace{0.215 M^{0.828}}_{\text{PFM}(M)} N^{\text{LRE}} \right]$$

M = platform mass [kg], N = units, cost in FY2025 €M.

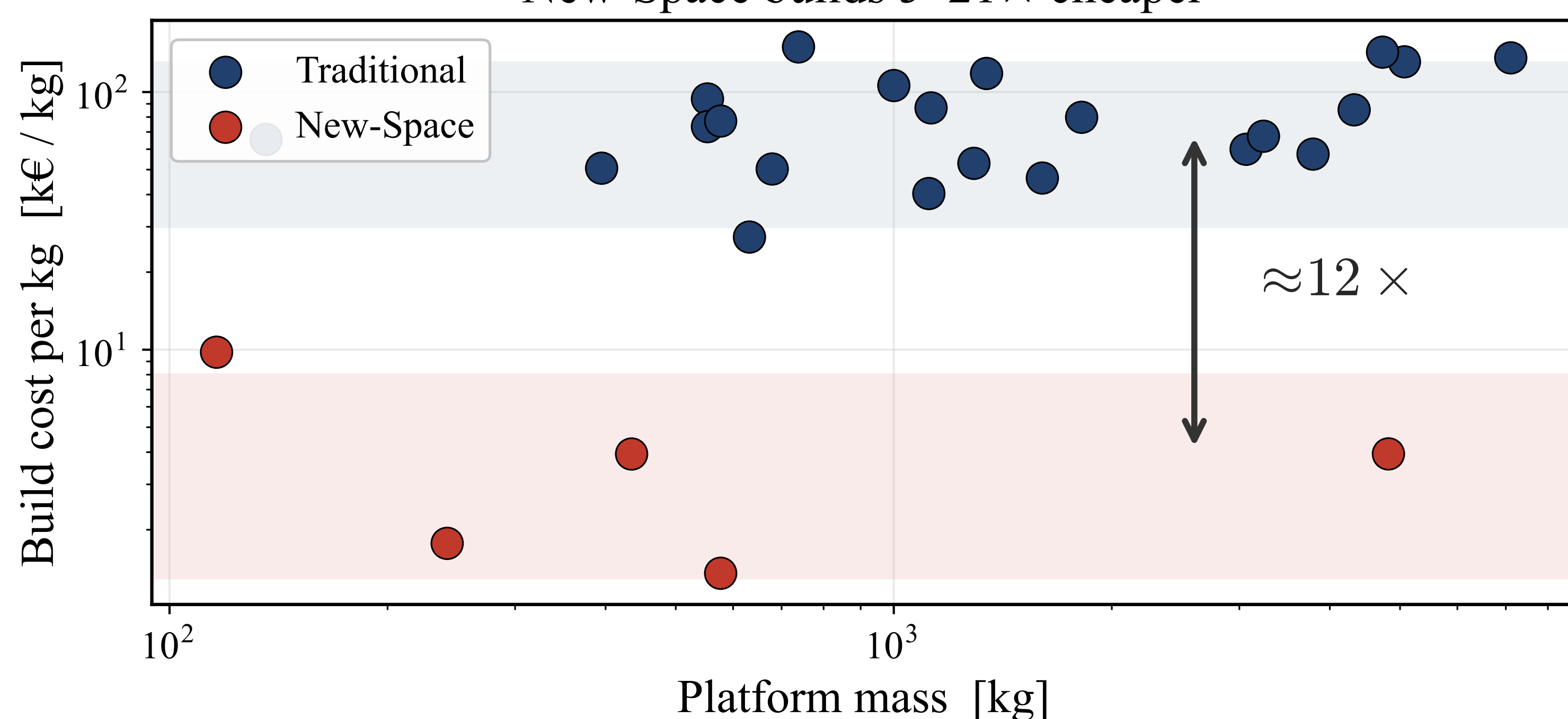
$\text{LRE} = 1 + \frac{\ln 0.9}{\ln 2} = 0.848$. $g = 1$ traditional, 0.118 New-Space.
 $F = 1$ Earth, 1.14 Mars.

The main results

0.92 cross-validated R^2
5–21× New-Space cheaper
 $\approx$ USCM8+QuickCost on traditional;
 but SMAD over-predicts New-Space $\sim 8\times$

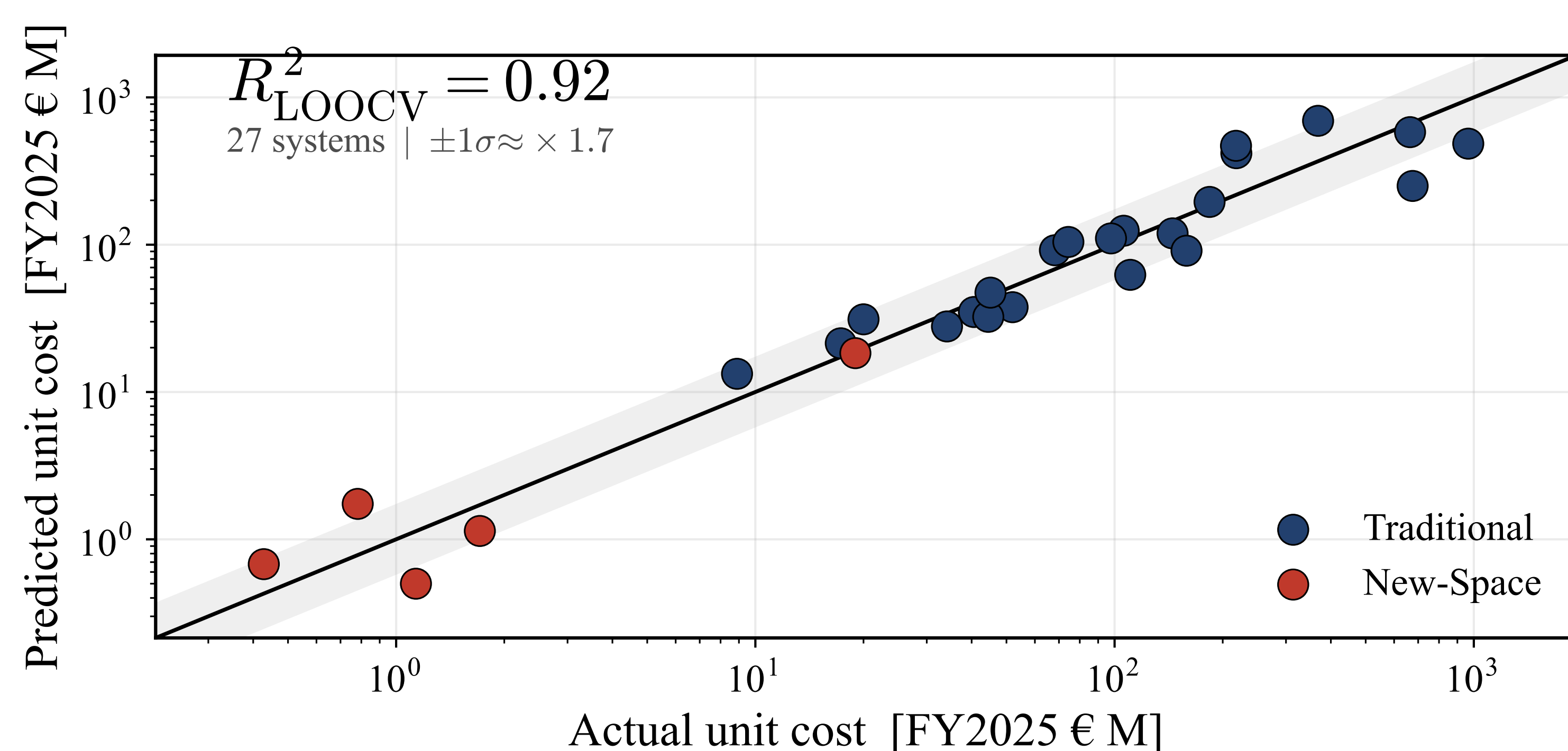
Cost-plus vs new-space economy

New-Space builds 5–21× cheaper



New-Space mass-produced builds are **5–21×** cheaper per kg, compared to traditional cost-plus bespoke systems.

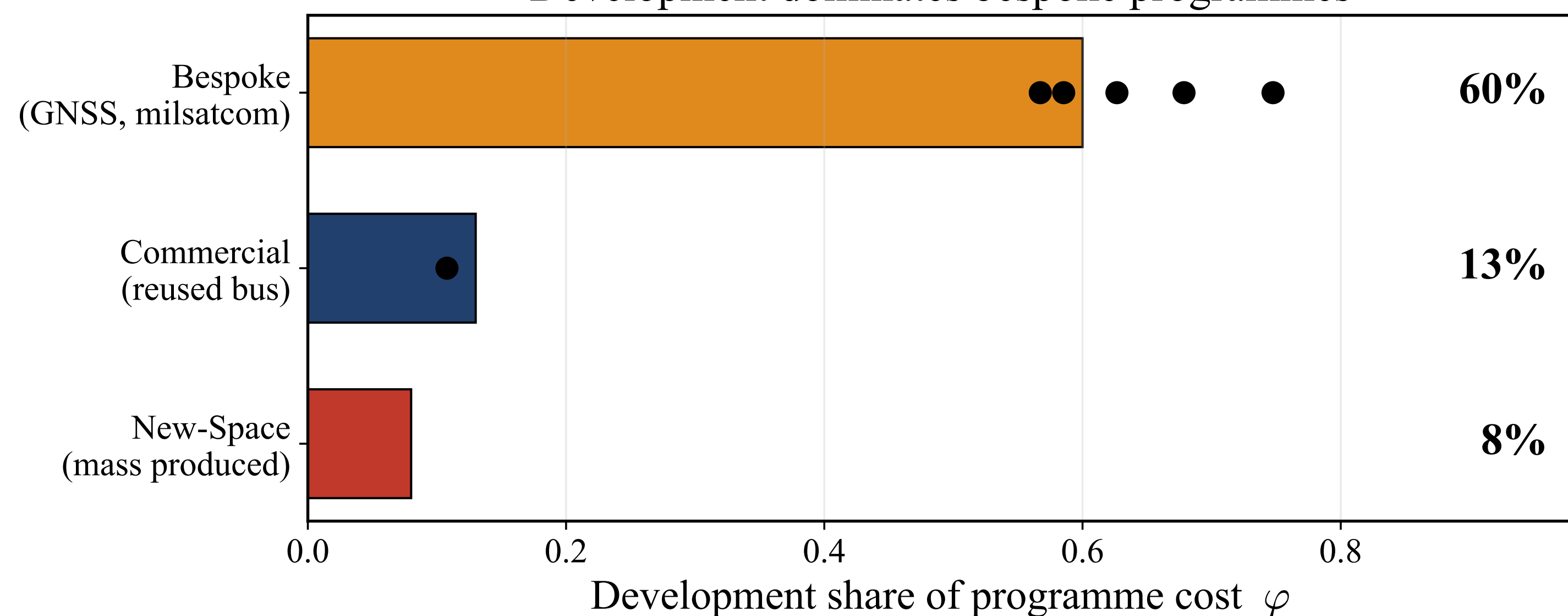
CER model validation



Leave-one-out cross-validation over 27 systems. The CER tracks unit cost across three orders of magnitude ($R^2 = 0.92$ pooled, 0.81 traditional-only).

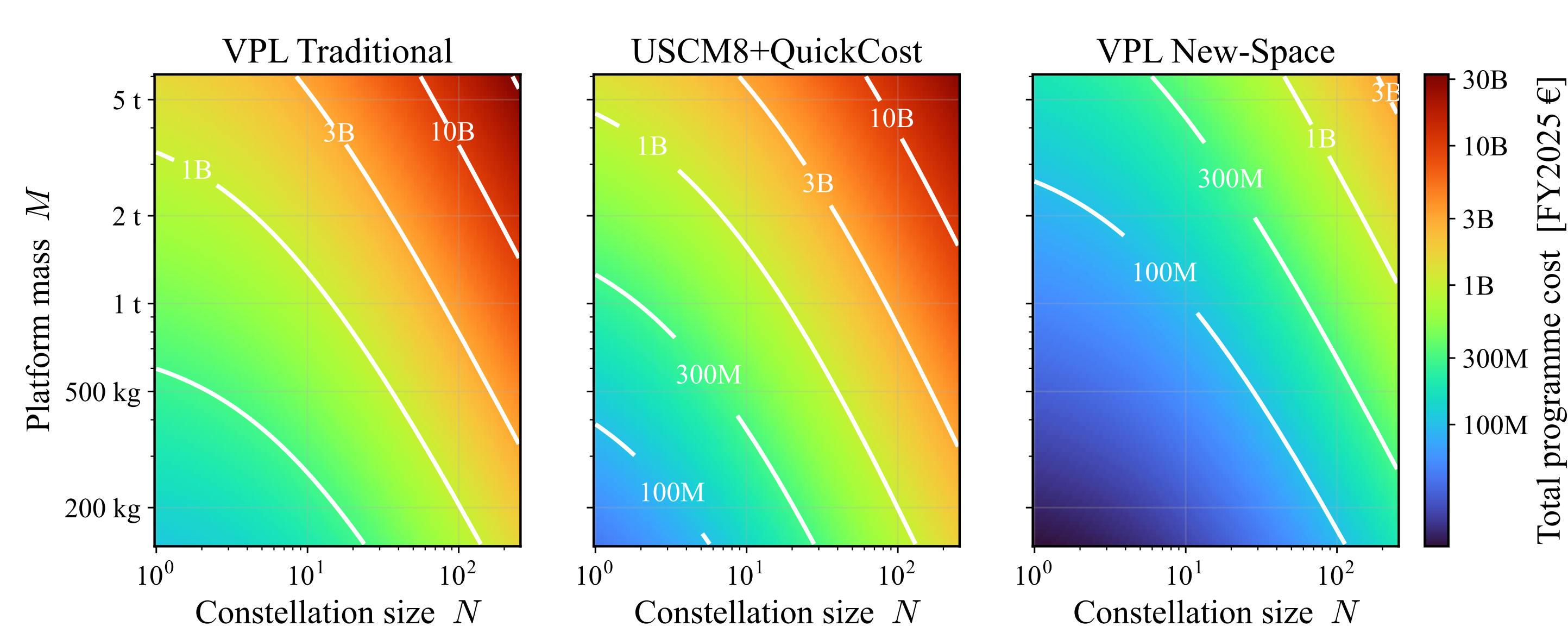
Development vs production costs

Development dominates bespoke programmes



The DEV/PFM split is fixed by disclosed programme budgets (US SARs, ESA, GAO).

Cost vs mass and constellation size



Iso-cost contours of $\text{TOTAL}(M, N)$. VPL Traditional tracks the SMAD benchmark, with VPL New-Space a full band below.

Planetary cost factor

A subsystem bottom-up estimate gives a Mars bus-cost increase of **10% ± 4.5%**, adopted as **F = 1.14** (Earth $F = 1$). This replaces SMAD/QuickCost's blanket 1.29 planetary factor. Sources: RAND TR-418; Jones (ICES 2015).

Operations cost

Mission operations use the SME-SMAD Table 11-28 model: **€159 M** for 31 satellites over 5 years. Drivers: flight and ground software maintenance (SLOC), 19 engineers + 4 technicians, facilities and ground-hardware maintenance.

Data and code



Github repository with dataset, sources, python model, reproduction notebook, poster and a detailed report.
github.com/cholevasm/vpl-cost-model